Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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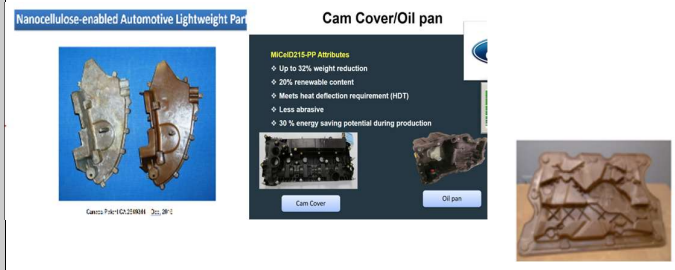
2. Technology Overview

Title of Proposed Technology	*A technology that enables hyper-performing cellulose nano-structured architecture for lighter, greener and safer mobility vehicle and building construction sectors*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Materials and manufacturing		*Advanced biocomposites*	
	Medium Category	Small Category	Medium Category	Small Category
	Energy efficient below freezing shear dispersion technology	*In-situ frictional heat dried cellulose nanofibrils composite*	*Functional composites*	*Flame retardant and anti-dripping functions*
Concept of Proposed Technology	Precision thermal management induced by a highly constrained hyper- shear generating containment in presence of a "onion-like multi-layered" natural materials, especially hydrophilic natural materials can lead to extraordinary anomalous expansion of liquid on freezing, resulting force amplification based on foundational fluid mechanics, leading to effective puncture of biological fibril architecture for facile release of micro- and nano-structured fibrils. This concept has been recently tested and validated in our lab, which could			

	lead to in-situ manufacturing of high performance rigid and flexible advanced composites with tunes functionalities. Such advanced bioproducts can be designed to tune functional properties for highly demanding performance applications ranging from automotive & building interior/exterior components to lighter drones and flexible displays.
Rationale for Joint R&D Project	This collaborative R&D projects calls for fundamental understanding of molecular conformational changes of natural and synthetic polymers and their thermodynamic mixes, i.e, composite, to exploit their translational merit in manufacturing automotive, building and electronic components that requires meeting stringent product specifications such as extraordinary thermo-mechanical and NVH (noise, vibration, harshness) tolerance behavior combined with low coefficient of thermal expansion yet resistant to flame, humidity and others. Here is some of the examples where CBBP technology successfully made in-road in North American and EU market:  CBBP Centre at the University of Toronto is a pioneer in natural fibre nanotechnology research and their industrial exploitation with more than 30 years of expertise, will be a key partner to explore new functional properties of cellulose micro-and nano-composites that is essential to meet structural and thermomechanical tuned for automotive, construction and electronic industries. Research at CBBP, in a lab scale or commercial scale (TRL 3-8) requires their cost-effective and marketable feature for commercial exploitation. The later skills are not readily available in an University set up. Therefore, collaborative partnership is essential for their ultimate commercial implementation. In this project, one of the first requirement is the capacity at scale for commercial volume demand. The later needs industry/SME partnership that are not only skilled to scale lab-based technology but also having resources and capacity to commercially produce and market it. Therefore, second essential reason to develop partnership is the supply chain development to manage the risk of target market deficiency. Many technologies do not go end up in consumer hand because of the lack of active involvement with the market driven enterprise. Therefore, this project bridges the gap between invention and innovation. By partnering with Korean SMEs, we are ceasing the opportunity to commercialize our technology and market them in Korean vast domestic automotive and construction market as well as reaching out to global opportunities using their marketing hub.

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Key Words	1. Cellulose plastic micro- and nano-composites 2. Rigid and flexible advanced bio-based substrates 3. Hyper-low thermal expansion coefficient 4. Tolerant to harsh environment, noise and vibration resistance 5. Functional attributes including fire retardancy, anti-fogging and anti-dripping

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (TRL3)	End of the Project	Level (TRL 6)
R&D Objectives	1. Functional and high strength cellulosic mirco- and nano-composites with ultra-low thermal expansion coefficient 2. Inducing flame-retardant properties in said composites with anti-dripping features 3. Develpoing an odorless composite design architecture that is resistant to NVH and humidity			
R&D Target Specifications	*Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters:* Target properties will depend on SME's target application. For Automotive sector with improved carbon footprint and to retain the target properties – we propose the following: 1. Maximum cellulose content in the composite. 2. The impact strength of composites containing cellulose is significantly low. Therefore, we target an impact strength increase rate of 20 to 25% over a benchmark cellulose composite that satisfies the target application ; 3. Tensile or flexural strength increase rate : We target an improvement of tensile strength and flexural strength over virgin polymer by 25 to 30% to meet target performance of a selected component 4. Flame retardancy rating (UL94 V0) : In partnership with SME, we shall target UL94 V0 spec for a targeted polymer matrix after maximizing cellulose content 5. Anti-dripping (no dripping during UL94 test) for the developed composite 6. thermal expansion coefficient. : reducing thermal expansion coefficient of the cellulose polymer composite by at least 2% over virgin polymer to meet the target performance. For building construction applications, we shall maximize the cellulose and bio-content to over 40% and for any application that relates to flexible cellulose composite our target will be less than 0.1% thermal expansion coefficient. The target application of any of those fields will be further defined once SME formally participates in this project.			
R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives* The key approach is to find best potential thermodynamic stability when ice crystals in cellulose fiber bundles with controlled moisture content are subjected to elongational shear in a contained chamber [1]. Fiber bundles are tightly entangled in a cellulose bundle. Overcoming the hydrogen bond energy by kinetically controlled vaporization of water induced by frictional thermal energy leads to practically water-free fibril generation and their *in-situ* dispersion in each thermoplastic polymer system. Here we propose optimizing defibrillation as a function of shear rate, temperature and cellulose content to achieve the level of dispersion of micro- and nano-fibrils of cellulose to achieve target mechanical performance, i.e., tensile and flexural stresses, flexural modulus and toughness of the composites. The optimization process will be carried out at University of Toronto's CBBP and/or Korean SME pilot facility at a TRL level 6, ensuring a production capacity of 50 kg/h. To achieve the thermodynamic stability of the mix of fibrillated cellulose and selective polymer			

in the above said contained chamber we propose to balance interfacial energy at the dynamically active polymer-fibril interface by doping reactive organic moieties that contain balanced hydrophobic and hydrophilic groups.

Once developed the optimal shear, time and temperature conditions – the composition will be further tuned to induce functional properties case to case basis depending on the target products defined by the SME parters. For example,

1. flame retardancy can be imparted to compositions by using large amounts of existing flame retardants, but as the flame-retardant content increases, the generation of toxic gases from the flame retardant increases. Additionally, lowering the flame-retardant content helps reduce unit costs, as flame retardants are usually expensive. Our research on functionalization will then drive by more focused active molecules which can effectively form ultra-thin layer of flame-retardant systems based on our earlier studies [2]. For example, we shall adopt a strategy to in-situ reactive grafting of N,P,B containing polymer, oligomer or monomers and their combinations; typically, a combination of polyimide, phosphonates and borates will be examined and optimized for in-situ graphing on the cellulose composites. Two approaches will be tested: 1. First we shall add them as a co-dopant in the defibrillation and in-situ high shear mixing in the kinetically controlled chamber as discussed above; 2. We shall dope this combination of chemical moieties during a second stage palletizations process using a twin screw extruder. We shall examine the UL 94 V0 flame retardancy and oxygen index as well as dripping properties.
2. Anti-dripping composite properties are important for automotive and construction industry. Since anti-drip agents are usually made of Teflon, scale-up work on developing and validating an anti-drip material in pilot scale that can replace Teflon seems necessary. Therefore, in yet another objective we shall also examine anti-dripping characteristics of the composite based on our previous experience on adding Nano fibrillated composition of marketed additives, developed in Korea that improves heat resistance by mineralizing the cellulose surface, to incorporate the flame retardant content and provides anti-drip properties. .
3. Our proposal also considers one other key challenge of automotive sector, i.e. odor of cellulose plastic composites. The odor generated from cellulose composites is thought to be due to thermal oxidation of cellulose during high-temperature plastic molding. We recently performed an in-depth study to reduce odors for automotive grade cellulose composites [3-5]. We shall adopt two approaches – first we shall study the effect of porous carbon coagent developed by us [4] to optimize the mechanical performance and odor resistant properties by incorporating 1 to 5% of our carbon additives. The mechanism of function has been described in our earlier report [4].

It is anticipated that incorporation of multiple co-agent as dopants might have effect on the interfacial properties of cellulose polymer composites; therefore, it is important for us to better understand the thermodynamic properties of the all co-dopants to retain thermodynamic stability of the composite interface in order to achieve best potential performance and to satisfy target properties for the potential application by the SME.

Milestones are as follows:

(1) Month 1-6 months : Recruitment of a PhD student in consultation with co-supervisor from MIE UofT; recruitment of PDF and 1 MS student; Complete literature review and refine plan of the project in consultation with SME

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	(2) Month 6-12 Examine and optimize cellulose fibril-polymer composite properties by enhancing interfacial adhesion. (PhD 1, PDf1) (3) Month 13-18 : Investigate the effect of copolymer and /or terpolymer coagent on the energy dissipation behavior of the composite composition above. Optimize the toughness properties of the composite without significantly sacrificing the mechanical performance (4) Month 19-24 : Investigate the effect of proprietary flame retardants and anti-dripping compounds derived by SME and CBBP,UofT group to achieve UL94 V0 performance by maintaining other performance properties a targeted application/s. Optimize their performance by carrying out statistical experimental design as a function of flame-retardant concentration and processing variables including temperature, shear and rheological properties. (5) Month 25-30 Examine the effect of CBBP's proprietary porous carbon material with bench mark commercial additives to maximize the odor suppression properties of the composite during and after manufacturing. Optimize overall composition of the target composite materials in relation to the final target properties specified by the SME. (6) Month 31-36 : develop scale-up strategy by refurbishing CBBP's proprietary equipment for the cellulose composite and demonstrate production of 300-500 kg of the production/h for coupon scaled up product testing in i380 T injection machine with SMEs Report/Invention disclosure/patent drafting as needed.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.* Research institute, CBBP at the University of Toronto will play key role in collaboration with Korean SME partner to carry out most part of the research and scale-up work that relates to unique process engineering related to the cellulose micro- and nanocomposite with tuned functional properties, design and process optimization, equipment building and scale up production of the based cellulose advanced composite materials. This institution will be two highly experienced professors will be cosupervising the project with their own time (3 to 7% of their time annually will be spent for this project depending on the student cosupervision and other project supervisory roles. Under the project operating umbrella CBBP will also engage their senior research associates and skilled technicians beyond students engagement to accelerate the scale-up developed of the proposed outcome. CBBP at the University of Toronto has heavily invested to develop infrastructure to carry out the proposed research which include proprietary defibrillation equipment, twin-screw scale extruders (80 and 100 mm); 380 T prototype injection machine for automotive parts validations and multi-layer profile extrusion technology scale up equipment to manufacturing production scale building and construction profiles based of cellulosics advanced biocomposites. The technical expertise available at University of Toronto is world-class and with able partnership with Korean SME and their expertise this culmination of knowledge bease will make this expert team as one of the world's best to compete globally. Prof. Sain has demonstrated tens of technology transfer in his 35 years of academic industry involvement and his supervision colleagues' are highly renowned for global technology transfer and value creations. When CBBP will primality focus of manufacturing and delivering at scale advanced cellulose polymer composites proposed in this program, additives and functional chemical will be tasked by the Korean SMEs to make the project a truly collaborative and win-win partnership. CBBP is also equipped with state-of-art analytical testing facilities required for this projects with over 10 Million dollar investment in past 15 years leading to establishment of advanced composite and flexible display device testing and characterization hub in Canada. Our Maldi-TOF-TOF, 3D Xray tomography, low-high temperature creep chamber for scaled automotive and construction components are second to none. Besides, we have weathering chambers, function properties such as anti-fogging , Oxygen index and other flammability testing equipment and odor qualification and testing are unique to our Centre's analytical portfolio. These post production performance and durability tests can be performed at CBBP in combination with the facility available at the partners SME locations.

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Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.* To successfully commercialize eco-friendly cellulose-reinforced plastics for the construction and automotive industries—where safety standards are paramount—it is essential to integrate UT Center's proprietary composite technology with advanced functionalizing solutions. While UT Center's composite technology provides a solid foundation for eco-friendly cellulose-reinforced plastics, customizing these materials for the construction and automotive sectors require further refinement in specialized functional areas. Specifically, achieving high-level flame retardancy and anti-dripping properties is critical to prevent fire spread and ensure structural integrity during thermal events. Furthermore, addressing the odor optimization of thermally degraded cellulose during high-temperature manufacturing is a key technical hurdle. We propose a collaborative framework where the South Korean partner develops localized, high-efficiency additive and deodorization technologies to refine these composites into market-ready, high-performance materials for automotive and building interior and exterior applications. University of Toronto CBBP group's core patented technology on micro fibrillation has been practiced by the industry. For example, Andersen Windows in US uses this technology for Window frame application where Prof. Sain acted as a consultant in mid nineties. Later, in Canada and USA, Prof. Sain led the commercialization of WPC microfibrillated composite industry which led to more than 80 companies in US and Canada to produce high performance exterior building components such as decks, fences, roofing and flooring by the year 2007. Subsequently China mass produced those products in large scale to dominate the market in low performance decking and fencing area whereas Prof. Sain advanced his innovation to pivot from low cost technology to hyper low temperature and high shear technology to develop new class of advanced cellulose technology and his new technology are now practiced by companies such as Greennano technologies when his first generation technology has been bought up by Domtar Inc. More lately Ford Motor Canada successfully developed high performance automotive engine components including battery tray, cam cover, oil pan, I pillar and many more. Mass scale production of this only can realize in partnership with a SME where automotive and construction domestic and globally competitive export market are readily accessible. South Korea is definitely a leading choice to implement these kind of technologies to capture global market.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.* In Korea, there have been attempts to reduce plastic usage by incorporating cellulose into plastics from an environmentally friendly perspective. However, interest has waned significantly due to a lack of functionality, including the physical properties necessary for use in automotive components. However, since there are commercial success stories in North America, I think Korea should also prepare. The highest impact of this collaborative work in the unique combination of proprietary ttechnologies of two global experts and export oriented countries in automotive and construction sectors tapping into global technology driven automotive clean technology market giving a significant cost and technology competitive benefits with their own technical talent expertise and abundant resources availability.

6. References

(1) M.Sain ; CA 2,549,844 ; Solid Phase Dispersion and Processing of micro- and nano-cellulose Fibres..

(2) J Feng et al. *ACS Sustainable Chemistry & Engineering*; Vol 5/Issue 9; July 25, 2017

(3)S. Amirabadi; Journal of Materials Science & Technology Volume 198, 1 November 2024, Pages 208-220

(4) B. Sun et al. Separation and Purification Technology Volume 354, Part 7, 19 February 2025, 129269

(5) Chemical Engineering Journal Advances, Volume 23, August 2025, 100765 “Synthesis of PTFE@PFAC core-shell composite particle as a highly efficient anti-dripping agent for PC”

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□ **Terms:**

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization